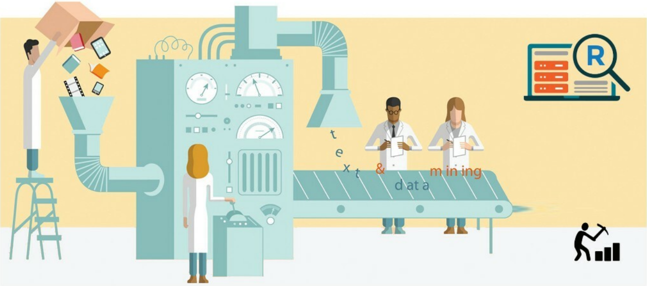
An Introduction to Text Mining with R

The vast quantity of data, textual or otherwise, that is generated every day has no value unless processed. Text mining, which involves algorithms of data mining, machine learning, statistics and natural language processing, attempts to extract some high quality, useful information from the text.



Text mining, in general, means finding some useful, high quality information from reams of text. More specifically, text mining is machine-supported analysis of text, which uses the algorithms of data mining, machine learning and statistics, along with natural language processing, to extract useful information. It covers a wide range of applications in areas such as social media monitoring, recommender systems, sentiment analysis, spam email classification, opinion mining, etc.

Whatever be the application, there are a few basic steps that are to be carried out in any text mining task. These steps include preprocessing of text, calculating the frequency of words appearing in the documents to discover the correlation between these words, and so on. R is an open source language and environment for statistical computing and graphics. It includes packages like tm, SnowballC, ggplot2 and wordcloud, which are used to carry out the earlier-mentioned steps in text processing.

Getting started

The first prerequisite is that R and R Studio need to be installed on your machine. R Studio is an integrated development environment (IDE) for R. The free open source versions of R Studio and R can be downloaded from their respective websites.

Once you have both R and R Studio on your machine, start R Studio and install the packages tm, SnowballC, ggplot2 and wordcloud, which are usually not installed by default. These packages can be installed by going to *Tools -> install packages* in R Studio.

Loading data

First, let us start by loading the data, which can be any text (or collection of texts, commonly referred to as a corpus) from which we want to extract useful information. To illustrate the process in this tutorial, we have used five text documents related to our favourite superhero, Batman!

For this step, first load the library tm and then use the function *Corpus()* to create a collection of documents. This loads our text documents residing in a particular folder into a corpus object.

```
library(tm)
docs <- Corpus(DirSource("path to your folder"))
docs
<<VCorpus>>
```